GROUP 4

JANUARY 12, 2026

CLASSICAL VS. DEEP LEARNING FACE DETECTION

FROM RULE-BASED LOGIC TO DATA-DRIVEN INTELLIGENCE

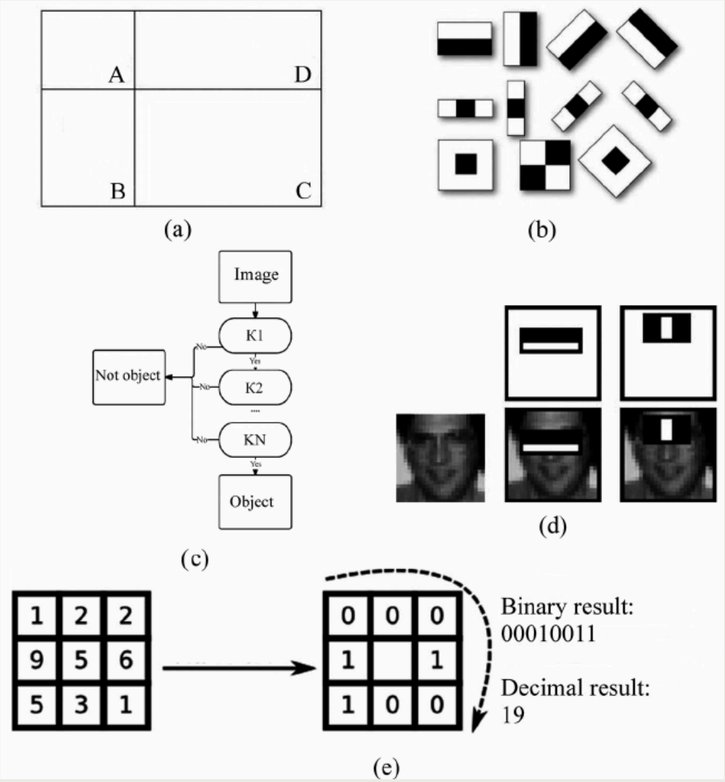
THE CLASSICAL APPROACH (VIOLA-JONES)

Assumes the image is mostly not a face. It attempts to eliminate background noise rapidly to isolate the face.

How It Works:

- Hand-Crafted Rules: Relies on human-defined contrasts (e.g., "The eye region must be darker than the cheek").
- Mechanism: Scans via a Sliding Window and a Cascade of Filters. If a section fails a rule, it is immediately discarded.

Outputs a Binary (Simple "Yes" or "No").



THE DEEP LEARNING APPROACH (YUNET)

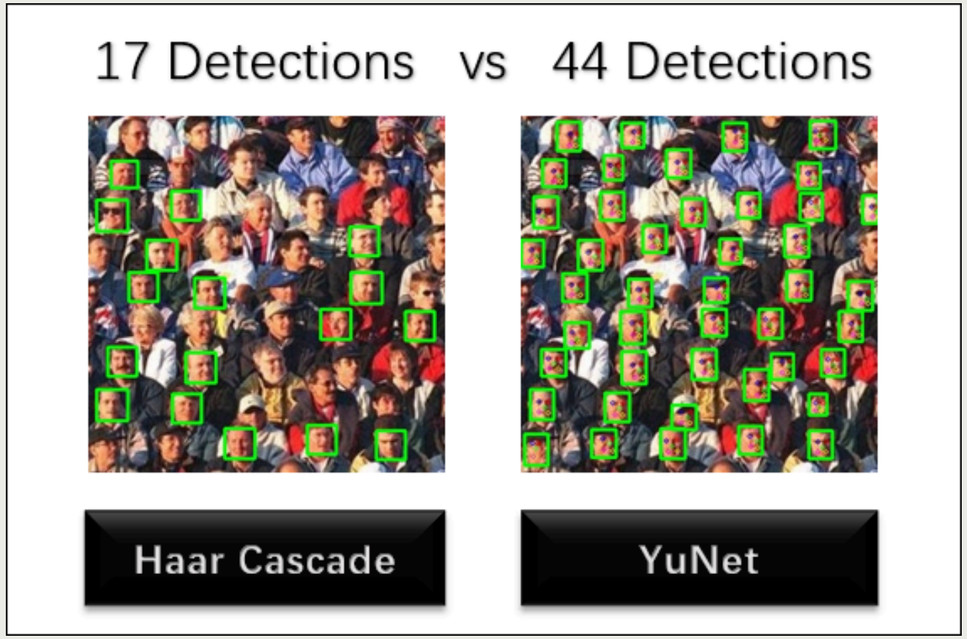
A specific, ultra-lightweight Convolutional Neural Network (CNN) designed to bring Deep Learning accuracy to standard CPUs and edge devices.

How It Works:

- Architecture: Uses Depthwise Separable Convolutions (a technique to reduce math operations) and a Feature Pyramid Network (FPN) to detect faces at different scales.
- Mechanism: Unlike the "sliding window," it views the image in a Single Shot, detecting small and large faces simultaneously in milliseconds.

Output is Rich Metadata which provides:

- Bounding Box (Location)
- Confidence Score (e.g., "99%")
- 5 Facial Landmarks (Eyes, Nose, Mouth corners).



KEY TAKEAWAYS: WHEN TO USE WHICH?

	Viola-Jones	**YuNet**
Logic Source	Human-Defined (Hand-crafted rules)	Data-Driven (Learned from millions of images)
Processing Style	Subtractive (Eliminates noise)	Pattern Matching (Finds structures)
Hardware Needs	Low (Standard CPU / Battery friendly)	High (often requires GPU)
Robustness	Fragile (Needs good lighting/frontal view)	High (Works in the wild, crowds, angles)
Best Use Case	Embedded systems, basic cameras, battery-constrained devices.	Security surveillance, facial recognition, AR filters, complex environments.

THANK YOU!